25. Probability

Year 4 · Mark Scheme — Supplementary Worksheet

Foundation — Answers

1. 1/6

2. 1/2

3. A — 0

4. 0.7

5. impossible, unlikely, evens, likely, certain

6. 50%

7. A — Impossible

8. 1

9. 1, 2, 3, 4, 5, 6

10. 2/5

11. A, C and E (0, 0.3 and 1 are all valid probabilities, which must be between 0 and 1; B is more than 1 and D is negative)

12. C — 1/2

13. 35%

14. 1, 2, 3, 4

15. 4/5

Standard — Answers

16. 2/5

17. 0.3

18. C — 0.4

19. HH, HT, TH, TT; P(exactly one head and one tail) = 1/2

20. 60%

21. C — 3/4

22. 4/9

23. A, B, C and D (P(rolling a 7)=0, P(even)=1/2, P(less than 7)=1 and P(rolling a 1)=1/6 are all true; E is false, since only 5 is a multiple of 5, so P = 1/6, not 1/3)

24. 3/5

25. 2/5

26. 18

27. A — 1/16 (1/4 × 1/4)

28. 2/9 (5/10 × 4/9)

29. 0.02

30. A, B, C and E (P(red)=2/5, P(not blue)=7/10, P(green)=P(blue)=3/10 and P(red or green)=0.7 are all true; D is false, since P(blue)=3/10=0.3, not 0.4)

31. 0.3

32. 1/3

33. 7/11

34. 1/2 (the multiples of 3 or 4 from 1–8 are 3, 4, 6 and 8, which is 4 out of 8 sections)

35. 5/8

Hard — Answers

36. 4 buttons (half of 24 = 12 red, a third of 24 = 8 blue, so green = 24 − 12 − 8 = 4)

37. 1/9 (the ways to make 9 are (3,6),(4,5),(5,4),(6,3) — 4 ways out of 36, which simplifies to 1/9)

38. 3/28 (3/8 × 2/7)

39. 1/5 (P(even on spinner) = 2/5, P(tails) = 1/2, so 2/5 × 1/2 = 1/5)

40. A, B, C and E (P(even)=1/2, P(multiple of 4)=3/12=1/4, P(greater than 10)=2/12=1/6 and P(prime)=5/12 are all true; D is false, since the square numbers 1, 4 and 9 give P=3/12=1/4, not 1/3)

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